In-Depth Telecom Churn Research Report: Economic Realities, Market Localizations, and Behavioral Machine Learning in Myanmar

Macroeconomics of Telecommunications: Core Metrics, Billing Models, and Churn Leverage

The commercial viability and operational scale of a telecommunications operator are governed by a complex matrix of usage metrics, acquisition economics, and customer retention dynamics. Average Revenue Per User (ARPU) is the fundamental financial benchmark used to measure the revenue-generating capacity of a subscriber base over a monthly billing cycle.[1]
Operators isolate top-line health by tracking blended, prepaid, and postpaid ARPU.[2] ARPU is mathematically defined as:

$$\text{ARPU} = \frac{\text{Total Service Revenue during Period } T}{\text{Average Number of Active Subscribers during Period } T}$$

Alongside ARPU, network utilization and user engagement are measured via Minutes of Use (MOU) and monthly data traffic per user, typically quantified in gigabytes.[1] MOU captures the voice network load, whereas data traffic tracking reflects the transition to a data-centric business model.[1]
As legacy voice revenues face ongoing dilution from Over-The-Top (OTT) messaging platforms like Telegram and Viber, data traffic has become the primary source of revenue growth.[1] This shift has turned the price per gigabyte and data consumption velocity into critical leading indicators of ARPU trajectories.[1]
A fundamental structural division exists between prepaid and postpaid billing paradigms. This division shapes how operators manage customer lifetime dynamics, distribution channels, and risk:

Metric / Dimension	Prepaid Subscription Model	Postpaid Subscription Model
Billing & Collection	Payments made in advance via top-up scratch cards, digital wallets, or retail	Monthly invoices generated in arrears based on service plans and overage

	agents.[1]	charges.[4]
Contractual Obligation	Zero contractual commitment; subscribers can switch providers instantly with no financial penalties.[6]	Fixed-term legal contracts (typically 12 to 24 months) with early-termination fees.[2]
Financial Risk Profile	Zero bad debt or credit risk; consumption is structurally capped by the prepaid balance.	Credit and default risks require structured scoring, billing audits, and collections management.
Distribution Strategy	Highly decentralized retail network with low-margin dealer commissions and virtual top-ups.[1]	Formal corporate outlets and authorized service centers handle identification and onboarding.[8]
Market Dominance	Highly dominant in developing markets (often exceeding 90% of the subscriber base).[3]	Predominant in developed, enterprise-centric markets where corporate billing is standard.[3]
Retention Dynamics	Highly volatile; prone to rapid brand switching and "silent churn" patterns.[6]	Structured retention based on contract lifecycles and early handset upgrade options.[2]

In emerging telecommunications markets, particularly across Asia, prepaid subscriptions dominate.[3] This makes managing the balance between Customer Acquisition Cost (CAC) and Customer Lifetime Value (LTV) a primary business focus. CAC represents the total financial outlay required to acquire a new active subscriber:

$$\text{CAC} = \frac{\text{Total Marketing} + \text{Sales Expenses} + \text{Dealer Commissions} + \text{SIM Kit Subsidies}}{\text{Total Newly Onboarded Subscribers during Period } T}$$

LTV represents the projected net profit generated by a single subscriber over the entire duration of their relationship with the operator, factoring in customer churn [2]:

$$\mathrm{LTV} = \frac{\mathrm{ARPU} \times \mathrm{Gross\ Margin\ Calculation}\ (\%)}{\mathrm{Monthly\ Churn\ Rate}}$$

The LTV-to-CAC ratio serves as a key measure of marketing efficiency and long-term business health; a ratio below 3:1 typically indicates unsustainable customer acquisition costs or high subscriber erosion.

In developing regions, annual prepaid churn rates often range from 20% to 70%, meaning some operators lose their entire equivalent subscriber base to churn within a single year.[5] Under these conditions, reducing customer attrition by even 1% can save a telecommunications company millions of dollars in marketing, dealer incentives, and SIM card printing costs.[2]

Because network capital expenditures (CapEx) and operational expenses (OpEx) remain high regardless of subscriber volume, retaining existing subscribers is five to ten times more cost-effective than acquiring new ones.[3] A highly stable subscriber base maximizes capacity utilization across cellular towers, yielding superior margins and allowing capital to be reinvested into advanced network technologies.

Regional Market Analysis: Competitor Landscape and Mobile FinTech Integration in Myanmar

The telecommunications landscape in Myanmar has experienced rapid transformation, evolving from a state-controlled monopoly before 2014 to a competitive four-operator market characterized by geopolitical shifts, rebranding, and local ownership consolidation.[11]

Competitor Landscape and Structural Profiles

Network Operator	Core Ownership Structure & Institutional backing	Active Subscriber Base & Revenue Indicators	Ecosystem Applications & Integrated Services
MPT (Myanma Posts and Telecommunications)	State-owned operator partnered with Japanese consortium KDDI and Sumitomo Corporation.[11]	Leads the market with approximately 43% to 44.5% market share [11], serving over 22 million active connections.[15]	**MPT4U**: Handles SIM registration updates, fiber migration management, and regional rewards.[1]
ATOM Myanmar (formerly Telenor)	Acquired by Lebanon's M1	Represents approximately 25%	**ATOM Store**: Offers integrated digital

	Group and local partners in 2022 after Telenor Group exited the market.[12]	to 28.4% market share [11], with about 17.23 million active connections.[15]	insurance, travel bookings, and in-app content packages.[1]
Mytel (Telecom International Myanmar)	Joint venture: Viettel (49%), Star High/MEC (28%), and Myanmar National Telecom Holding (23%).[16]	Represents ~14% market share.[11] Peak base reached 13 million, with USD 2 billion in accumulated revenue (2018–2022).[17]	**MyID**: High-engagement super-app with 23.4 million installations by 2025.[1]
U9 (formerly Ooredoo Myanmar)	Sold to Singapore-based Nine Communications (Link Family Office and U Nyan Win) for USD 576 million.[13] Rebranded as U9 in 2025.[19]	Represents approximately 13% market share.[11] Transitioned existing users to U9-branded assets.[19]	**U9 App**: Replaced the Ooredoo app, offering data cashbacks and integrated wallet recharges.[19]

This competitive landscape has been shaped by significant geopolitical shifts. Following political transitions in 2021, Norway's Telenor Group exited the country, selling its mobile business to M1 Group.[12] This unit was subsequently rebranded as ATOM Myanmar.[12] Before its exit, Telenor experienced substantial customer erosion, losing over 2 million subscribers in 2018 alone.[15]
This decline was driven by aggressive price competition from Mytel, which was granted a temporary exemption from regulatory price floors to capture early market share.[15] Consequently, ATOM's monthly voice ARPU declined to NOK 24 (approximately USD 2.85) in late 2018.[15]
Mytel, operated as a joint venture with Vietnam's Ministry of National Defence (Viettel), grew rapidly through heavy fiber deployment, building over 30,000 kilometers of fiber-optic cable.[11]
This infrastructure enabled nationwide 4G services.[11] However, Mytel has faced significant consumer boycotts tied to its ownership structure.[16] In the first half of 2021, these boycotts led to a subscriber loss of approximately 2 million users and a 30% reduction in its sales workforce.[16]
The qatar-headquartered Ooredoo Group also exited the market, finalizing the sale of Ooredoo Myanmar to Nine Communications Pte. Ltd. for an enterprise value of USD 576 million and equity consideration of USD 162 million.[13] Under Chief Executive Officer Daw Caroline Yin Yin Htay, the operator launched a complete rebranding transition to **U9** on September 20, 2025.[19]

The transition maintained physical and eSIM card compatibility while launching a redesigned self-care application and a promotional campaign offering up to 9GB of free data to users texting "Hello U9" to 9999.[19]
Myanmar's financial infrastructure has largely bypassed traditional retail banking, positioning mobile money wallets as the primary payment and distribution layer for the country's digital economy.[1] Mobile operators utilize these wallets to facilitate airtime recharges, data package purchases, and subscription payments.

- **KBZPay:** Developed by Kanbawza Bank, KBZPay recorded over 2.1 billion transactions in 2024, serving as a key platform for digital recharges and financial transactions.[1]
- **Wave Money:** Supported by an extensive network of 61,000 agents, Wave Money provides 90% geographic coverage across Myanmar, making it highly effective for reaching rural communities.[1]
- **MytelPay:** Operated by Mytel, this wallet is integrated directly into the MyID super-app ecosystem, lowering barriers to digital transactions.[1]

To offset the decline of legacy SMS and voice revenues, operators have introduced "Super-Apps" like MyID (which recorded 23.4 million installations by 2025) and the ATOM Store.[1] These applications bundle self-care account management with lifestyle features, including video streaming, music platforms, and gaming micro-transactions.[1] By integrating mobile wallet payments directly into these super-apps, operators can capture micro-fees, improve user engagement, and increase blended ARPU through targeted, in-app data promotions.[1]

Regulatory and Infrastructure-Induced Churn Mechanisms

Telecommunications operators in Myanmar navigate highly complex operational environments, where regulatory compliance and infrastructure vulnerabilities serve as major catalysts for involuntary customer churn.[1]

SIM Card Registration Regulations

The Post & Telecommunications Department (PTD) enforces strict SIM card registration rules, requiring every mobile connection to match identity data held by the Ministry of Immigration and Population.[8] Under these mandates, subscribers must register their SIM cards using their National Registration Card (NRC) details, providing clear front and back photographs of the physical card, along with their full name, township code, and registration type in the Burmese language.[8]

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The standard NRC format—such as "12/MaBaTa(N)123456"—must match the physical ID.[20] If registration records do not match government databases, operators send warning SMS alerts under sender IDs like "MPTSimAlert".[8] This starts a strict 30-day grace period for correct re-registration.[8]

Subscribers who fail to update their registration details through self-care apps (such as the MPT4U or U9 apps) face immediate outgoing service suspension, blocking voice calls, data access, and mobile wallet OTP messages.[8] This is followed by permanent SIM deactivation and recycling of the number.[8]

Central Equipment Identity Register (CEIR) Mandates

Adding to these regulatory pressures, the government has introduced a Central Equipment Identity Register (CEIR) system, requiring all mobile devices to register their International Mobile Equipment Identity (IMEI) numbers.[21] Under rules in effect by March 31, 2026, newly imported mobile devices must register their IMEIs and pay associated customs duties to avoid being blacklisted.[21]

Devices that fail to comply with these tax and registration mandates are blocked from connecting to any local cellular network, leaving them unable to access voice, SMS, or data services.[21] This regulatory intervention has caused a sharp rise in involuntary churn as uncertified or smuggled handsets are disconnected.[21]

Grid Failures and Operating Cost Volatility

In addition to regulatory challenges, operators must manage severe infrastructure vulnerabilities. Myanmar suffers from chronic grid instability, with summer electricity shortages forcing rotating power outages that can last up to 20 hours daily.[1] These blackouts require operators to rely heavily on backup diesel generators to maintain cellular tower uptime.[1]

This operational dependence introduces significant cost volatility:

- **High Diesel Dependence:** Out of more than 23,000 active cellular towers in the country, fewer than 1,300 are equipped with solar-hybrid backup systems, leaving the vast majority dependent on diesel fuel.[1]
- **Fuel Supply Disruptions:** Local fuel prices are highly sensitive to border trade restrictions—such as Thailand's fuel supply cuts to border towns in early 2025—which can cause regional diesel prices to spike.[24]
- **Hyper-Inflation and Currency Risk:** Triple-digit diesel inflation, combined with currency depreciation (with the kyat falling to MMK 4,300 per USD), has inflated network operating expenses.[1]
- **Infrastructure Security:** Ongoing regional conflicts add to these costs; attacks on infrastructure damaged or destroyed 305 cellular towers between 2021 and 2023, driving up insurance premiums and limiting new network investment.[1]

These severe operational challenges directly affect customer retention. As fuel costs rise and power outages persist, operators struggle to maintain consistent quality of service (QoS).[1] Increased network congestion, dropped voice calls, and slow data speeds frustrate subscribers, driving voluntary churn as users seek operators with more stable regional coverage.[5]

Technical Feature Engineering, Silent Churn, and Machine Learning Modeling

In prepaid-dominated markets, predicting customer attrition requires a shift from tracking formal contract cancellations to analyzing subtle behavioral indicators.[6] Because prepaid users do not have contractual obligations, they rarely cancel their subscriptions formally.[6]

Instead, they exhibit "silent churn" or "usage compression," leaving their primary SIM card inactive or using it only as a backup while moving their active voice and data traffic to a competitor's secondary SIM.[9]

To identify these at-risk subscribers, operators analyze telecommunications datasets to map technical features to real-world behavioral changes.[4]

Telecom Churn Technical Feature Dictionary

Technical Feature Name	Business Meaning	Core Behavioral Indicator

rev_Mean	Mean monthly revenue (total charge amount).[4]	Directly measures customer spend; rapid declines indicate silent churn.[9]
mou_Mean	Mean monthly minutes of use.[4]	Measures voice network utilization; declines signal shifting usage.[3]
totmrc_Mean	Mean total monthly recurring charge.[4]	Tracks baseline service commitment; drop-offs indicate plan downgrades.
da_Mean	Mean directory assisted calls.[4]	Reflects user reliance on assistance services.
ovrmou_Mean	Mean overage minutes of use.[4]	Measures usage exceeding standard plan allowances.
ovrrev_Mean	Mean overage revenue.[4]	Measures high-margin overage billing.
vceovr_Mean	Mean voice overage revenue.[4]	Isolates high-value voice overages.
datovr_Mean	Mean data overage revenue.[4]	Measures data overages; drops suggest the user is rationing data or has a secondary SIM.[9]
roam_Mean	Mean roaming calls.[4]	Tracks travel usage; sudden drops indicate a reduction in mobility or local SIM usage.[3]
change_mou	Percentage change in monthly minutes of use vs. previous three-month average.[4]	Crucial indicator of silent churn; sustained negative values show voice usage compression.[3]
change_rev	Percentage change in monthly revenue vs.	Measures revenue compression; helps identify

	previous three-month average.[4]	users who are reducing their spend.[9]
eqpdays	Age of current mobile handset in days.[25]	Identifies device obsolescence; high values suggest the user is at risk of churning due to handset limits.[25]
drop_vce_Mean	Mean dropped voice calls.[4]	Measures quality of service; spikes predict voluntary churn due to poor network performance.[5]
drop_dat_Mean	Mean dropped data sessions.[25]	Measures data reliability; critical for identifying users frustrated by network quality.[5]

Analyzing these features in combination allows operators to build automated systems to flag silent churn before the subscriber is lost. For example, a user with stable historical spending who exhibits a sharp decline in usage metrics—such as deep negative shifts in change_mou and change_rev, combined with a drop in data overages (datovr_Mean) and an increase in dropped call rates (drop_vce_Mean)—presents a clear signature of silent churn.[3]
This profile suggests the user is actively migrating their traffic to a competitor's network due to localized network quality issues, even though their primary SIM remains registered and technically active.[5]

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Historically, predictive models have evolved from simple linear classifications to advanced machine learning architectures capable of identifying non-linear patterns across large subscriber bases.[7] Academic and industry research shows that ensemble methods, particularly XGBoost and LightGBM, achieve high predictive accuracy (ranging from 94% to 95.74%), outperforming Support Vector Machines (SVM) and K-Nearest Neighbors (kNN).[3]
Furthermore, Gradient Boosting models achieve balanced F1-scores of around 0.6098 on highly imbalanced telecommunications datasets, making them effective for targeted retention efforts.[28]
To maintain model accuracy, operators must align predictions with broader digital adoption trends. DataReportal metrics trace Myanmar's rapid digital growth alongside recent speed volatility:

- **Internet Users:** At the end of 2025, Myanmar’s active internet users reached 39.8 million, representing a 72.5% penetration rate.[30] This reflects a significant increase from 24.11 million users in early 2024.[31]
- **Mobile Connections:** Active connections totaled 62.5 million (equivalent to 114% of the population).[30] This represents a 2.0% decline (-1.3 million connections) from late 2024, reflecting subscriber consolidation and SIM cleanups.[30]
- **Broadband Adoption:** Approximately 96.8% of active connections are classified as broadband (3G, 4G, or 5G).[30]
- **Quality of Service (QoS) Shifts:** In early 2025, median mobile download speeds dropped significantly by 78.2% to 5.09 Mbps, down from 23.31 Mbps in 2024.[31] This decline was driven by network congestion and localized power outages.[1]

These shifts in network quality directly impact customer behavior. As mobile internet speeds

degrade, subscribers are more likely to experience network issues.[5]
If operators do not monitor these quality drops alongside usage trends, they risk missing early silent churn signals from users who are quietly moving their traffic to other networks.[5]

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To prevent majority-class bias in highly imbalanced datasets (such as a 73% retained to 27% churned ratio), data science teams use Synthetic Minority Over-sampling Technique (SMOTE) to balance training records.[9] Missing metrics are resolved using K-Nearest Neighbors (kNN) imputation to preserve underlying behavioral patterns.[3]
Additionally, operators can optimize the predictive window based on their business model.[34] While short-term predictions (e.g., 7 to 30 days) yield higher accuracy, expanding the forecast horizon (e.g., to 90 or 180 days) provides marketing teams with a wider window to design and execute retention campaigns.[34]

Strategic Synthesis: Segmented Retention and

Localized Campaigns

To maximize return on investment, telecommunications operators must distinguish between subscribers who can be saved through targeted marketing interventions and those whose churn is driven by factors outside the operator's control.[6]

Retainable vs. Non-Retainable Subscriber Segments

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- **Retainable Segments:** This group includes subscribers experiencing usage compression due to outdated devices (indicated by high eqpdays values) or those exhibiting early signs of silent churn (such as minor declines in change_mou or change_rev) due to temporary network issues.[6] These users can be retained through device upgrade financing or targeted service promotions.[2]
- **Non-Retainable Segments:** This group includes subscribers facing involuntary churn due to regulatory non-compliance, such as unverified NRC credentials or blacklisted IMEIs under the CEIR system.[8] It also includes users in conflict-affected regions experiencing complete, long-term network outages due to destroyed towers.[1] In these cases, marketing promotions or discounts are ineffective, and operators should conserve their retention budgets.[6]

Localized Intervention Campaigns

For the retainable segment, operators can deploy targeted, automated intervention campaigns that leverage Myanmar's unique digital wallet networks and device preferences.[1]

Campaign A: Digital Ecosystem and Lifestyle Bundles

- **Target Segment:** Mobile users showing moderate declines in data overages (datovr_Mean) and usage metrics (change_mou), but who are highly active on super-apps and digital wallets.[1]
- **Mechanism:** Operators can partner with major mobile money networks like KBZPay or Wave Money to offer targeted data promotions and cashbacks.[1] For example, when a user tops up their account via KBZPay, the system can automatically trigger a personalized data bundle matched to their usage history.[1]
- **Ecosystem Integration:** These data packages can be bundled with gaming tokens, music streaming, or micro-insurance products through super-apps like MyID or the U9 App.[1] By offering integrated lifestyle value rather than simple discounts, operators can counter OTT voice and SMS cannibalization, increase screen time, and improve subscriber stickiness.[1]

Campaign B: Handset Modernization and Device Financing

- **Target Segment:** High-value subscribers with high handset equipment age (e.g., eqpdays > 500 days) who are using older, 3G-only devices and experiencing high dropped data rates (drop_dat_Mean).[5]
- **Mechanism:** To prevent these users from migrating to competitors' 4G-Advanced or 5G networks, operators can launch targeted device upgrade campaigns.[1]
- **Execution:** Operators can partner with leading device manufacturers like Xiaomi or Oppo to offer affordable 4G/5G smartphones, financed through micro-loans integrated directly into KBZPay, Wave Money, or MytelPay.[1] To secure these loans and protect against credit risk, the financing can be tied directly to the device's registered IMEI number under the CEIR framework.[1] This strategy helps subscribers upgrade their hardware, improves their network experience, and transition volatile prepaid users into sticky, long-term hybrid or postpaid subscription models.[1]

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